ROHIT PRAJAPATI

AI/ML Engineer | Generative AI & LLMs | RAG | Python

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PROFESSIONAL SUMMARY

B.Tech Computer Science graduate with hands-on experience building AI and machine learning applications using Python, LLM APIs, RAG pipelines, embeddings, vector databases, LangChain, FastAPI, Streamlit, and Docker. Built end-to-end projects involving machine learning, deep learning, NLP, document retrieval, LLM integration, REST APIs, and application deployment. Strong foundation in Python, machine learning, deep learning, and Generative AI, with a strong interest in understanding AI systems beyond hosted APIs, including retrieval, inference, evaluation, and production applications. Quick learner who enjoys experimenting with new technologies, debugging technical problems, and building practical AI solutions.

TECHNICAL SKILLS

Programming & Data: Python · SQL · Pandas · NumPy · EDA · Regex · Feature Engineering

Machine Learning: Scikit-learn · Supervised Learning · Unsupervised Learning · Ensemble Methods · Model Evaluation

Deep Learning & NLP: TensorFlow · ANN · CNN · RNN · LSTM · Transformers · Tokenization · Embeddings · Text Classification · Sentiment Analysis

Generative AI: LLMs · RAG · Prompt Engineering · LangChain · LangGraph · AI Agents · LLM APIs · Vector Databases · FAISS

Backend & Deployment: FastAPI · Flask · Streamlit · Docker · REST APIs

Tools: GitHub · VS Code · Jupyter Notebook

PROJECTS

Generative AI Q&A Application — RAG Pipeline

Python · LangChain · FAISS · OpenAI · Gemini · Groq · Streamlit

• Built an end-to-end Retrieval-Augmented Generation application for document-based question answering using LLM APIs, embeddings, FAISS vector search, and LangChain.

• Implemented document processing, text chunking, embedding generation, similarity-based retrieval, and context-aware response generation.

• Integrated multiple LLM providers through APIs and developed an interactive Streamlit application for real-time document Q&A.

• Worked with retrieval and prompt configurations to understand how retrieved context affects generated responses.

California House Rent Prediction — End-to-End ML Application

Python · Scikit-learn · Pandas · NumPy · FastAPI · Streamlit · Docker

• Performed exploratory data analysis, data preprocessing, feature engineering, model training, and evaluation using Python and Scikit-learn.

• Built a FastAPI backend exposing REST endpoints for model predictions and connected it with a Streamlit interface.

• Containerized the application using Docker to create a portable deployment environment.

• Evaluated the machine learning model using R^2 and Mean Squared Error.

Deep Learning Image Classifier — CNN + Transfer Learning

Python · TensorFlow · Streamlit

• Built a deep learning image classification application using a CNN and transfer learning.

• Trained and evaluated the model on image data and integrated the trained model into a Streamlit application.

• Worked with model training and prediction workflows to understand practical deep learning application development.

INTERNSHIP EXPERIENCE

Data Science & AIML Intern — YBI Foundation

Jun 2024 – Jul 2024

• Completed a hands-on Data Science and AI/ML internship involving exploratory data analysis, supervised machine learning, and model-building techniques.

• Worked with real-world datasets under mentor guidance and applied machine learning concepts through practical tasks.

• Developed practical understanding of data preprocessing, model development, and evaluation.

Credential ID: YOWAUK57UMYC

EDUCATION

B.Tech — Computer Science Engineering

SAGE University, Indore, Madhya Pradesh · 2021 – 2025

CERTIFICATION

Full-Stack Data Science & AI Specialization

Industry Certification Program — Completed

Python · Machine Learning · Deep Learning · NLP · Generative AI · Agentic AI · MLOps